



**THE UNIVERSITY OF QUEENSLAND
SCHOOL OF CIVIL ENGINEERING
CIVL4333 Advanced Concrete Design
Assignment 2**

A simply-supported prestressed concrete beam with a total span of 22 m is designed to carry a permanent uniformly distributed superimposed dead load of 3 kN/m and live load of 12 kN/m in addition to its self-weight. The beam is post-tensioned by two parabolic cables, each of which consists of seven strands of nominal diameter 15.2 mm. The maximum eccentricities at mid-span of these cables are indicated in Fig 1. The cable profiles are **symmetric** about the mid-span as shown in Fig 1. Each cable is stressed initially to 70% of the characteristic breaking load sequentially. The material and section properties are as follows:

Concrete

- Characteristic concrete strength = 32 MPa (28th day); 25 MPa (at transfer)
- Young's Modulus = 30,100 MPa (28th day); 26100 MPa (at transfer)
- Density of concrete = 24 kN/m³

Prestressing strands nominal diameter 15.2 mm

- Characteristic breaking load = 250 kN
- Nominal area = 143 mm²
- Young's Modulus of strand $E_p = 190,000$ MPa
- Relaxation factor = 3% of characteristic breaking load

Cross section properties

- Area of concrete $A_g = 363,000$ mm²
- Second moment of area (moment of inertia) $I = 62 \times 10^9$ mm⁴
- Centroid = 650 mm measured from bottom
- Hypothetical thickness $t_h = 177$ mm
- Area of non-prestressing steel $A_s = 1,250$ mm²
- Mean annual temperature = 20°C

Others

- Friction coefficient $\mu = 0.2$
- Wobbling coefficient $\beta_p = 0.016$ rad/m
- Anchorage draw-in $\delta_{ad} = 6$ mm

- Determine the total short-term loss of prestress at transfer at mid-span.
- Determine the ultimate loss of prestress at service after 30 years at mid-span. Assume a temperate inland environment condition.

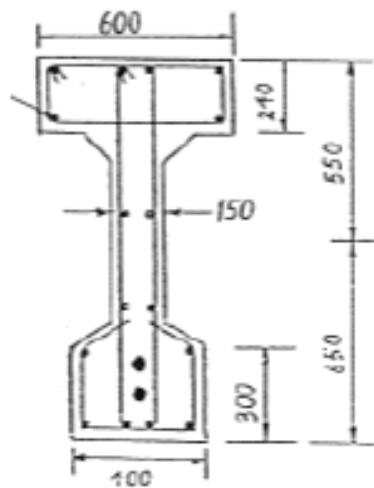
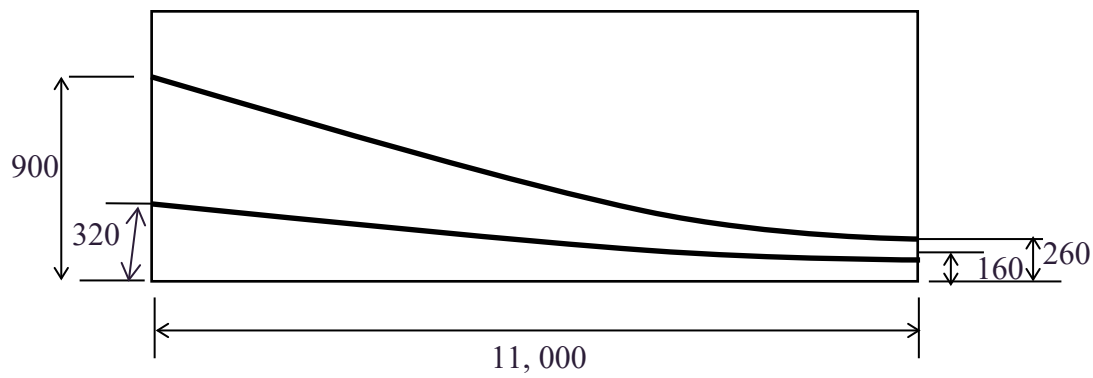


Fig 1 Elevation of beam showing prestressing steel profile (half span only) and cross section at midspan

END OF ASSIGNMENT 2